

Northern Gannet Range Prediction

Interactive Species Distribution Modelling with FastAPI & Streamlit

STAT 418 · June 2026

Containerised ML deployment on Google Cloud Run

Project Overview & Motivation

What are Northern Gannets?

- Seabirds native to the North Atlantic
- Strong indicators of ocean health & prey availability
- Range shifting northward with climate change

Project Goal

- Predict Northern Gannet habitat range by year & month
- Deploy an interactive tool accessible via the web

Tech Stack

Data

GBIF API → 305k occurrence records

Model

XGBoost Regressor

API

FastAPI · Uvicorn

Frontend

Streamlit · Cartopy

Deploy

Podman → DockerHub → Google Cloud Run



Data Collection & Preprocessing

Data Source

- GBIF (Global Biodiversity Information Facility)
- 305,581 occurrence records - *Morus bassanus*
- Fields: lat, lon, year, month

Preprocessing

- Filtered to Northeast Atlantic (lat 45-75, lon -30-30)
- Removed records with missing coordinates/dates
- Centered year: $\text{year_centered} = \text{year} - 1980$

Features → Target

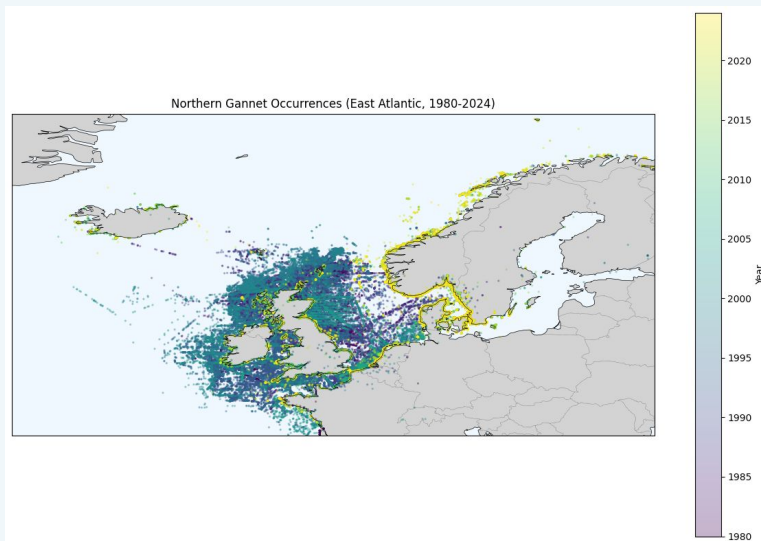
- decimalLongitude
 - year_centered
 - month_sin / month_cos
- **predict decimalLatitude**

Cyclical Month Encoding

Month is encoded as sin/cos to preserve its cyclical nature. A raw integer (1-12) would imply December (12) is far from January (1), introducing false structure.

$$\text{month_sin} = \sin(2\pi \times m/12)$$

$$\text{month_cos} = \cos(2\pi \times m/12)$$



Model Development & Evaluation

Model	MAE (°)	R ²	Notes
Baseline	3.28	-0.35	Predicts mean lat always
Linear Regression	4.16	-0.64	Worse than baseline - non-linear problem
Random Forest	1.97	-0.11	Low MAE but negative R ² - overfitting
XGBoost ✓	2.14	+0.22	Only model with positive R ² - selected

2.14° MAE

~240 km error with only
4 features — reasonable

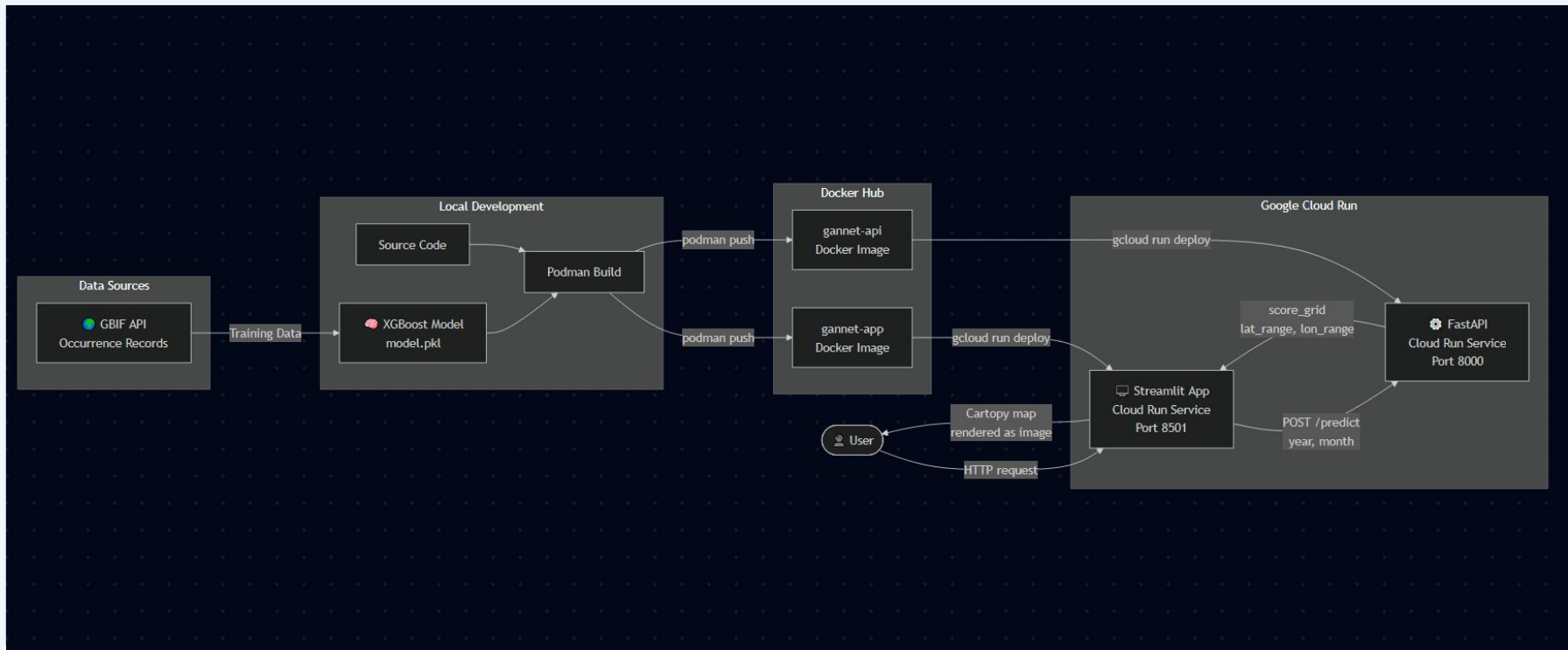
Only +R²

XGBoost is the sole model
that explains variance

Gaussian viz

Point predictions converted
to probability surface

Solution Architecture



GBIF API
Occurrence data source

XGBoost + pkl
Trained model artifact

FastAPI / CR
POST /predict endpoint

Streamlit / CR
Interactive web UI

Docker Hub
Image registry

Application Features & User Interface



Year & Month Sliders

1970-2030 range including future projections. Month displayed by name.



Cartopy Map Overlay

Probability heatmap on PlateCarree projection with coastlines & borders.



Compare Mode

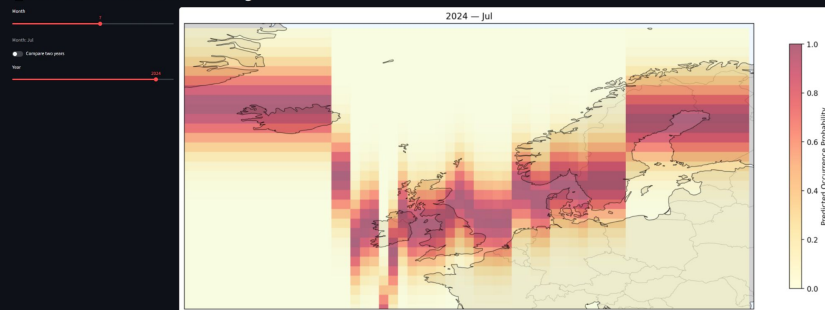
Side-by-side two-year view with custom user labels (e.g. Historical / Projected).



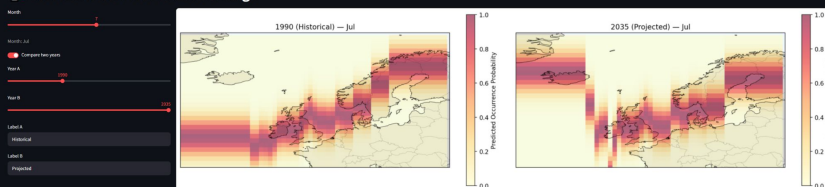
Cached API Calls

st.cache_data - repeated year/month combos don't re-hit the FastAPI endpoint.

Northern Gannet Predicted Range



Northern Gannet Predicted Range



Challenges, Solutions & Lessons Learned

01

Challenge

Data limitations with only spatial and temporal features

Solution

Compared XGBoost with baseline and simpler models, was able to learn some signal compared to other models

02

Challenge

North American data had significant outliers from uncategorized sources

Solution

Changed range to North Atlantic region bounding Europe

03

Challenge

Predicting gridded probability scores using all features didn't perform well

Solution

Stuck with predicting latitude using other features, had usable performance and visualization

04

Challenge

Plotly geo traces don't support heatmaps on geographic projections

Solution

Switched to st.pyplot() with Cartopy